Interpretable Federated Kolmogorov–Arnold Surrogates for Autonomous and Evolutive Resource Optimization over Non-Stationary LEO Links

Phuc Hao Do[1] and Truong Duy Dinh[2]

Abstract—**Low Earth orbit (LEO) networks must repeatedly solve resource-allocation problems whose optima have no closed form and that drift as satellites pass and interference regimes switch. We study an autonomous and evolutive approach in which a federated Kolmogorov–Arnold network (KAN) learns the *solution map* of an energy-efficiency transmit-power optimization, mapping a link state to its optimal power online. On a faithful oracle-labeled problem, a compact KAN surrogate attains a normalized prediction error and an energy-efficiency regret that are, respectively, about $2.3\times$ and $1.7\times$ lower than a parameter-matched multilayer perceptron (MLP) at equal parameter budget (Wilcoxon $p < 0.01$), while a linear predictor fails entirely, confirming the map is strongly nonlinear. Crucially, the additive spline structure of the KAN lets us *read off closed-form design rules* from the trained model, a transparency an MLP cannot provide. A contextual-bandit controller sparsifies the federated updates, halving the uplink volume while retaining the accuracy advantage over the MLP, tracing a favorable communication–accuracy frontier. We further report a transparent negative result: growing the spline grid online (self-evolution) does not improve over a well-sized static grid in this federated streaming regime. The study positions interpretable KAN surrogates as a practical building block for autonomous optimization in non-stationary networked-AI systems.**

Index Terms—**Kolmogorov–Arnold networks, federated learning, learning to optimize, LEO satellite networks, energy efficiency, interpretability, non-stationary environments.**

I. Introduction

NEXT-generation non-terrestrial networks built on LEO constellations operate under fast, non-stationary conditions: the channel gain sweeps with elevation across a pass, and handovers between satellites and beams switch the interference regime on a timescale of seconds [1], [2]. Each terminal must continually re-solve resource-allocation problems such as energy-efficient power control, whose optimizers have no closed form and must be recomputed as conditions drift. Solving these from scratch online is costly, and the time-varying feasible region defeats precomputed lookups [3].

A learning-to-optimize view replaces the per-instance solver with a surrogate that maps a problem state directly to its optimal decision [3]. For this to be trustworthy in a networked-AI system it should be (i) accurate on the deterministic solution map, (ii) compact and communication-light when trained in a federated manner across many terminals [4], [5], and (iii) interpretable, so operators can audit the learned policy. Multilayer perceptrons meet (i)–(ii) only partially and fail (iii) outright.

We argue that Kolmogorov–Arnold networks (KANs) [6] are a natural fit. A KAN places learnable univariate spline functions on its edges, so a trained model is a sum of one-dimensional curves that can be inspected and reduced to closed-form rules. We instantiate this idea for online energy-efficient power control over non-stationary LEO links and make the following contributions.

- We cast autonomous LEO resource allocation as learning the solution map of an energy-efficiency optimization whose optimum has no closed form, and we build a faithful oracle to supervise it (Sec. III).
- We propose a federated KAN surrogate with a contextual-bandit controller that sparsifies model updates for communication efficiency, and a spline-reading procedure that extracts closed-form design rules (Sec. IV).
- Over ten seeds the KAN surrogate lowers prediction error by about $2.3\times$ and energy-efficiency regret by about $1.7\times$ versus a parameter-matched MLP (Wilcoxon $p < 0.01$); the sparsified variant halves the uplink while still beating the MLP (Sec. VI).
- We report a transparent negative result: online grid self-evolution does not beat a well-sized static grid here, clarifying where "evolutive" capacity growth does and does not help (Sec. VI, VII).

II. Related Work

Learning to optimize trains networks to emulate optimization solvers for wireless resource management [3]; prior art almost exclusively uses black-box MLPs/CNNs and does not address interpretability or non-stationary deployment. **KANs** [6] and efficient implementations [7] have shown strong function-fitting and symbolic-extraction ability, but their use inside networked optimization is nascent. **Communication-efficient federated learning** reduces uplink via sparsification and quantization [4], [8], [5]; we add a learned controller that adapts sparsity to a budget. **Concept-drift adaptation** [9] motivates online tracking; here the drift is the physical regime switching of an LEO pass [1].

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III. System Model and Problem Formulation

A. Energy-efficient power control

Consider a set $\mathcal{N}$ of gateway/UAV terminals served by LEO backhaul. A served link in state $s = (g, I, p_0)$ has channel gain g, interference I, and a circuit/energy term p_0. With transmit power p, the energy efficiency is

$$\mathrm{EE}(p; s) = \frac{\log_2\bigl(1 + gp/(1+I)\bigr)}{p + p_0} \quad [\text{bits/Hz/J}], \tag{1}$$

which is quasi-concave in p with a unique maximizer $p^\star(s) = \arg\max_{0 \le p \le P_{\max}} \mathrm{EE}(p; s)$ [10]. The stationarity condition is transcendental, so $p^\star(\cdot)$ has no closed form, yet it is a smooth, deterministic function of s, the ideal regime for a spline-based surrogate.

B. Non-stationarity over an LEO pass

Along a pass the elevation sweep moves the achievable gain g, while satellite handovers switch the interference level $I_n(t)$ in a piecewise-constant fashion [1]; visibility windows gate which terminals can participate. The operative region of the state space, and thus the relevant slice of $p^\star(\cdot)$, therefore drift in time, so the surrogate must be learned and refreshed online rather than fit once.

C. Problem

Let $\hat{p}_\theta$ be a shared surrogate with parameters θ and let $m_{n,t} \in \{0,1\}^{|\theta|}$ be the uplink mask of terminal n at slot t. We minimize the time-averaged energy-efficiency regret subject to a per-slot uplink budget $B(t)$:

$$\begin{aligned} \min_{\theta, \{m_{n,t}\}} \; & \frac{1}{T} \sum_{t=1}^{T} \mathbb{E}_{s \sim \mathcal{D}_t} [\mathrm{EE}(p^\star(s); s) - \mathrm{EE}(\hat{p}_\theta(s); s)] \\ \text{s.t.} \; & \textstyle\sum_n \|m_{n,t}\|_0 \, \beta \le B(t), \quad \forall t, \end{aligned} \tag{2}$$

with θ aggregated across $\mathcal{N}$ and $\mathcal{D}_t$ the (drifting) state distribution. Problem (2) is non-convex (surrogate training), coupled through the shared θ and the budget, and time-varying through $\mathcal{D}_t$.

IV. Proposed Method

A. KAN surrogate

We use a two-layer KAN regressor $\hat{p}_\theta : \mathbb{R}^3 \to \mathbb{R}$ with B-spline edges [6], [7]. Each layer computes $y_o = \sum_i \bigl(w^b_{oi}\,\mathrm{SiLU}(x_i) + \sum_k c_{oik}\, B_k(x_i)\bigr)$, a sum of univariate functions of the inputs, which is what makes the trained model readable.

B. Federated online learning with budgeted sparsification

At each slot the visible terminals draw states at their current regime, label them with the oracle $p^\star$, take local SGD steps, and send sparsified updates that are FedAvg-aggregated [4]. A contextual-bandit controller [11] observes a context (regime indicator, budget price, recent error) and selects the keep-fraction of the top-magnitude update entries, trading communication against accuracy under $B(t)$. A drift detector [12] on the surrogate-error stream flags regime switches for optional refresh.

TABLE I
Mean over ten seeds. Lower is better except R^2.

Method	NMSE	EE regret	Uplink (Mb)	rule R^2	#par
fedkan_opt (ours)	0.079	0.038	**4.8**	0.81	360
kan_full	**0.058**	**0.022**	9.0	0.77	360
kan_evolve (ablation)	0.080	0.038	5.2	0.81	404
mlp	0.135	0.038	9.0	–	361
mlp_prox	0.170	0.038	9.0	–	361
linear	2.292	0.352	0.1	–	4

C. Reading closed-form rules

Because the first-layer response to each input is a single univariate curve, we probe each feature in isolation and fit a small library of forms (linear, quadratic, logarithmic, square-root, reciprocal), keeping the best by R^2. This yields human-readable rules such as "$p^\star$ grows near-linearly with the circuit term p_0 but nonlinearly with channel gain g," which no MLP exposes.

D. Evolutive capacity (ablation)

We also test *self-evolution*: on a drift flag the controller may grow the spline grid, adding knots to refine the surrogate. As Sec. VI shows, this does not help here; we therefore keep a static grid in the proposed method and report evolution as an ablation.

V. Experimental Setup

LEO geometry is generated from elevation-dependent path loss; interference follows piecewise-constant regimes with handover switches; ten terminals participate over four passes. Each slot draws 128 oracle-labeled states. All surrogates are parameter-matched (≈ 360 parameters). We compare: **fedkan_opt** (proposed, bandit-sparsified KAN), **kan_full** (KAN, dense updates), **kan_evolve** (proposed + grid self-evolution), **mlp** and **mlp_prox** [8] (parameter-matched MLPs), and **linear**. Metrics: normalized MSE (NMSE) of $\hat{p}$ versus $p^\star$, energy-efficiency regret, total uplink bits, extracted-rule fidelity R^2, and parameters; means over ten seeds with Wilcoxon signed-rank tests.

VI. Results

A. Accuracy and communication

Table I summarizes the study. The KAN surrogates clearly dominate the MLPs: `kan_full` attains NMSE 0.058 versus the MLP's 0.135 (2.3× lower, Wilcoxon $p = 0.004$, lower in 9/10 seeds) and regret 0.022 versus 0.038 ($p = 0.002$, 10/10). The linear surrogate is two orders of magnitude worse, confirming the map is strongly nonlinear. The proposed `fedkan_opt` cuts uplink volume in half (4.8 vs. 9.0 Mb, $p = 0.002$) while still beating the MLP on NMSE ($p = 0.027$), i.e. it dominates the MLP on both accuracy and communication. Fig. 1 shows the resulting communication–accuracy frontier.

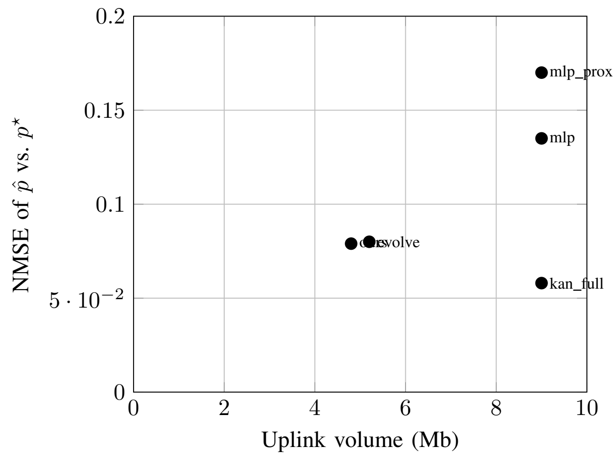

Fig. 1. Communication–accuracy frontier. The proposed bandit-sparsified KAN (ours) dominates both MLP variants and halves the uplink of the dense KAN.

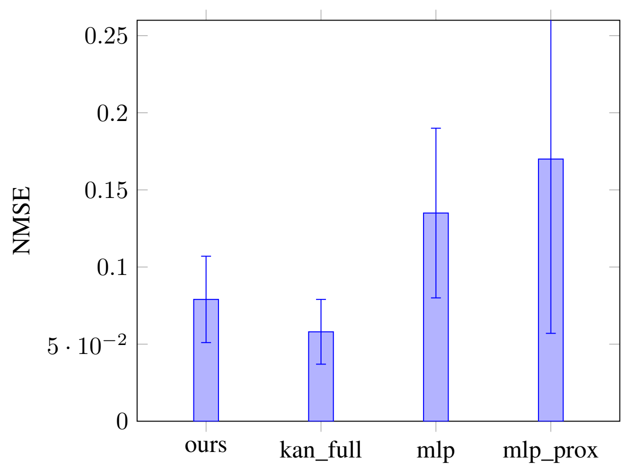

Fig. 2. NMSE (mean±std, ten seeds). Both KAN variants are well below the MLPs.

B. Online behavior under drift

Fig. 3 tracks the global NMSE across slots as interference regimes switch. The KAN surrogates stay consistently below the MLP throughout the non-stationary horizon, recovering quickly after regime changes.

C. Interpretability

Fig. 4 plots the learned univariate responses of `kan_full` to each state variable together with the best-fit closed form (mean fidelity $R^2 = 0.74$). The optimal power depends near-linearly on the circuit term p_0 ($R^2 = 0.99$) and on interference I ($R^2 = 0.88$), but nonlinearly on channel gain g ($R^2 = 0.36$ for a single basis), which the spline captures. Such rules are a direct, auditable output of the model.

D. Does self-evolution help?

No. `kan_evolve` matches neither `kan_full` (NMSE 0.080 vs. 0.058, $p = 0.014$, evolution lower in only $2/10$ seeds) nor improves on the proposed static-grid

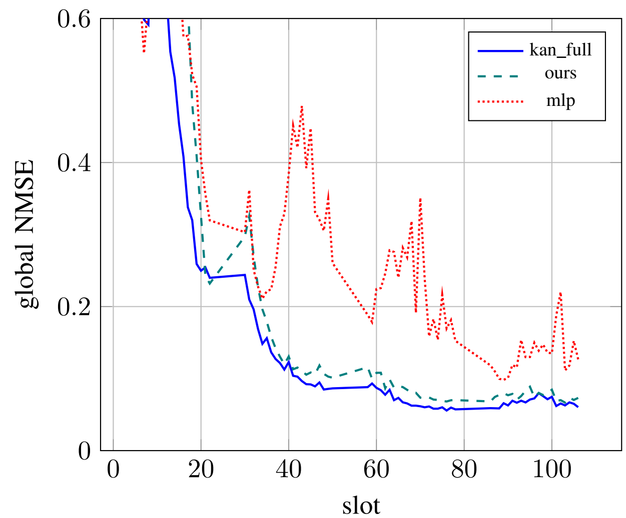

Fig. 3. Global NMSE over the non-stationary horizon (representative seed).

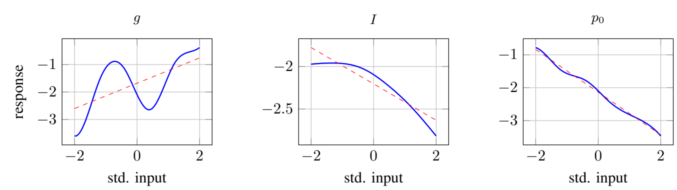

Fig. 4. Extracted univariate rules (solid) and best-fit closed form (dashed). The p_0 and I dependences are near-linear; g is nonlinear and carried by the spline.

`fedkan_opt`, while using more parameters. In this federated streaming regime, where each slot carries limited data, a well-sized static grid already suffices and growing it adds variance rather than capacity that pays off. We report this as a boundary of the "evolutive capacity" idea.

VII. Discussion and Limitations

The energy-efficiency objective is flat near its optimum, so even imperfect power predictions incur small utility regret; this is why the MLP's regret is close to the KAN's despite a much larger prediction error, and why we report both metrics. The interpretability gain is qualitative but concrete: the KAN yields auditable rules, the MLP does not. Our oracle and channel model are simulation-based; closing the loop on measured LEO traces and extending the decision to joint power and rate/sub-band allocation are natural next steps. Finally, the negative self-evolution result is specific to this low-data-per-slot federated setting and may differ when per-task data is abundant.

VIII. Conclusion

We presented an interpretable federated KAN surrogate for autonomous resource optimization over non-stationary LEO links. It learns the energy-efficiency solution map about 2.3× more accurately than a parameter-matched MLP, halves the uplink through learned sparsification while still beating the MLP, and yields closed-form design rules. A transparent ablation shows that online grid self-evolution does not help here. Interpretable KAN surrogates are a promising component for autonomous and evolutive optimization in networked-AI systems.

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